

SOFTWARE ENGINEERING

LAB 1

Name : Pallavi J

SRN : PES1UG24CS313

Section : 5F

Problem Statement : Digital Campus Library Reservation Gateway

=> Functional Requirements :

Description	Priority	Acceptance Criteria	Rationale
The system shall allow student members to search the physical library catalog using an ISBN or book title.	High	For a valid ISBN or title query, the system shall display all matching catalog records.	Enables students to quickly locate required library resources.
The system shall allow student members to place a hold request on a currently borrowed book.	High	When a student selects an eligible borrowed book and submits a hold request, the system shall add the student to the book's reservation queue and display the queue position.	Provides the core reservation functionality for unavailable books.
The system shall maintain reservation queues in first-in-first-out (FIFO) order.	High	When a reserved copy becomes available, the system shall assign it to the student at the front of the queue.	Ensures reservations are handled fairly according to request order.
The system shall	High	When a borrowed	Ensures students are

notify the next eligible student when a reserved book becomes available.		book is returned and becomes available, the system shall send a hold-availability notification to the student at the front of the queue.	informed promptly when their reserved book is available.
The system shall record and display overdue fine transactions associated with student library accounts.	Medium	When an overdue transaction is recorded, the system shall calculate and display the applicable fine in the student's account.	Supports accurate tracking of overdue library transactions.

=> Non Functional Requirements :

Description	Priority	Acceptance Criteria	Rationale
Performance : The system shall return paginated catalog search results within 150 ms for a catalog containing up to 50,000 titles under simulated peak load.	High	In performance testing with 50,000 catalog titles, at least 95% of valid search requests shall return results within 150 ms.	Ensures fast catalog access during high-demand periods.
Security : The system shall restrict reservation and fine-management operations to authenticated users with the appropriate permissions.	High	An unauthenticated or unauthorized user attempting a restricted operation shall be denied access.	Protects student information and prevents unauthorized transactions.
Availability : The system shall maintain a minimum availability of 99.5% during scheduled	High	Monthly monitoring shall show system availability of at least 99.5% during scheduled operating	Ensures students and librarians can access essential library services reliably.

operating hours each month.		hours.	
Usability : The system shall provide clear confirmation and error messages for reservation and catalog operations.	Medium	In usability testing, at least 90% of test users shall correctly identify whether their requested operation succeeded or failed based on the displayed message.	Reduces user confusion and helps students understand the outcome of their actions.
Data Integrity : The system shall preserve the FIFO order of reservation queues when reservations are added, cancelled, or fulfilled.	High	After adding, cancelling, or fulfilling reservations, the queue order shall match the expected FIFO order in 100% of test cases.	Prevents unfair reservation ordering and maintains accurate queue information.